

## PROGRAM 1: FACTORIAL OF A NUMBER

```
fact=function(num){  
  if (num==0||num==1){  
    return(1)  
  }else{  
    return(num*fact(num-1))  
  }  
}  
  
cat("Enter a Number:")  
num<-readLines(con="stdin",n=1)  
num=as.numeric(num)  
print(num)  
result=fact(num)  
cat("Factorial:",result,'\n')
```

OUTPUT (for input: 5):

```
[1] 5  
Factorial: 120
```

## PROGRAM 2: PALINDROME CHECKER

```
palin<-function(result){
og<-result
rev<-0
while(result>0){
a<-result%%10
rev<-(rev*10)+a
result<-result%%10
}
if(og==rev){
cat("It is a pallindrome\n")
}else{cat("It is not a pallindrome\n")}
}

cat("Enter a Number:")
num<-readLines(con="stdin",n=1)
num=as.numeric(num)
print(num)
palin(num)
```

OUTPUT (for input: 121):

```
[1] 121
It is a pallindrome
```

## PROGRAM 3: LINEAR REGRESSION

```
data(mtcars)
model=lm(mpg~wt,data=mtcars)

print(summary(model))

newcars<-data.frame(wt=c(0.2,0.3,0.5))
predicted=predict(model,newdata=newcars)
results=data.frame(weight1000lbs=newcars$wt,
                    PREDICTED_MG=predicted)

print(results)

plot(mtcars$wt,mtcars$mpg,
     main="MPG VS WEIGHT",
     xlab="MPG", ylab="WEIGHT",
     pch=14, col="blue")

abline(model,col="red",lwd=2)
```

# PROGRAM 3: LINEAR REGRESSION - OUTPUT

Call:

```
lm(formula = mpg ~ wt, data = mtcars)
```

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 37.2851  | 1.8776     | 19.858  | < 2e-16 ***  |
| wt          | -5.3445  | 0.5591     | -9.559  | 1.29e-10 *** |

Residual standard error: 2.949 on 30 degrees

Multiple R-squared: 0.7528

Adjusted R-squared: 0.7446

F-statistic: 91.38 on 1 and 30 DF

p-value: 1.294e-10

Predictions:

|   | weight1000lbs | PREDICTED_MG |
|---|---------------|--------------|
| 1 | 0.2           | 38.017       |
| 2 | 0.3           | 36.677       |
| 3 | 0.5           | 33.996       |

[Plot generated: MPG vs WEIGHT with regression line]

## PROGRAM 4: LOGISTIC REGRESSION

```
data<-read.csv("student_data.csv")
print(data)
model<-glm(Result ~ Hours,data=data,
            family=binomial)

cat("Logistic Regression Model Summary:\n")
print(summary(model))
new_data<-data.frame(Hours=5.5)

probability<-predict(model,new_data,
                     type="response")

cat("\n Predicted Probability of passing:",
    probability, "\n")

prediction<-ifelse(probability>=0.5,1,0)

cat("Predicted Class(0=Fail,1=Pass):",
    prediction, "\n")

class<-ifelse(prediction==1,"Pass","Fail")
cat("Final Prediction:",class, "\n")
```

## PROGRAM 4: LOGISTIC REGRESSION - OUTPUT

Call:

```
glm(formula = Result ~ Hours, family = binomial,  
     data = data)
```

Deviance Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -1.8624 | -0.6486 | 0.2667 | 0.6644 | 1.4755 |

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |   |
|-------------|----------|------------|---------|----------|---|
| (Intercept) | -4.0777  | 1.7610     | -2.315  | 0.0206   | * |
| Hours       | 0.8240   | 0.3738     | 2.204   | 0.0275   | * |

Residual deviance: 38.23 on 18 degrees of freedom

AIC: 42.23

Predicted Probability of passing: 0.7342

Predicted Class(0=Fail,1=Pass): 1

Final Prediction: Pass

## PROGRAM 5: VARIANCE, COVARIANCE & CORRELATION

```
data(mtcars)

# Selecting relevant columns
data_subset <- mtcars[, c("mpg", "wt", "hp")]

cat("Variance:\n")
print(var(data_subset))

cat("\n\nCovariance Matrix:\n")
print(cov(data_subset))

cat("\n\nCorrelation Matrix:\n")
print(cor(data_subset))
```

# PROGRAM 5: VARIANCE, COVARIANCE & CORRELATION

## - OUTPUT

Variance:

|  | mpg       | wt       | hp          |
|--|-----------|----------|-------------|
|  | 36.324103 | 9.769596 | 4700.866935 |

Covariance Matrix:

|     | mpg         | wt         | hp          |
|-----|-------------|------------|-------------|
| mpg | 36.324103   | -5.1169341 | -320.732185 |
| wt  | -5.116934   | 9.7695968  | 44.192610   |
| hp  | -320.732185 | 44.1926098 | 4700.866935 |

Correlation Matrix:

|     | mpg        | wt         | hp         |
|-----|------------|------------|------------|
| mpg | 1.0000000  | -0.8676594 | -0.7761684 |
| wt  | -0.8676594 | 1.0000000  | 0.6587479  |
| hp  | -0.7761684 | 0.6587479  | 1.0000000  |